

SAVITRIBAI PHULE PUNE UNIVERSITY



A FINAL PROJECT REPORT

ON

Image Geolocation and Hazard-Aware Response

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING

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Abstract

Image Geolocation and Hazard-Aware Response is a multimodal image understanding system that combines place-aware visual reasoning with risk-aware decision support. The system accepts an image from the user, analyzes visual and contextual cues using a vision-capable multimodal large language model, and returns a structured response containing summary, possible location, risk level, confidence, entities, rationale, and recommended actions. For normal landmark or place images, the system provides informational actions such as learn-more and map lookup. For hazardous or suspicious scenes, the system shows report-oriented safety actions only after explicit user confirmation.

The project is implemented as a Python Flask backend, SQLite persistence layer, and React-based web interface. Gemini 2.5 Flash is used as the primary multimodal model, while deterministic post-processing logic validates model output, normalizes confidence, applies hazard keyword rules, and maps risk states to user-facing actions. The system stores sessions, messages, model outputs, actions, and incident records for auditability. The evaluation uses project run logs containing 22 analyzed runs across 14 sessions. Low-risk and high-risk outputs each occurred eight times, uncertain outputs occurred six times, and the average confidence was 0.623. Ten user-confirmed incident records and twelve learn-more actions were logged. The results show that structured multimodal reasoning can support both geospatial understanding and controlled hazard response, while preserving human verification for safety-sensitive actions.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Image Geolocation and Hazard-Aware Response is an intelligent image analysis system designed to identify possible geographical locations from images while also assessing potential hazards present in the scene. The system helps users understand visual content by providing meaningful information about the surroundings, important landmarks, environmental conditions, and possible risk factors detected within an uploaded image.

The system accepts an image as input and performs comprehensive scene interpretation. Based on the visual information available in the image, it generates a structured response that includes a brief description of the scene, probable location details, detected objects or landmarks, confidence level of the prediction, identified risk category, and appropriate recommendations for the user.

In addition to location estimation, the system is capable of recognizing situations that may indicate danger, emergency, or suspicious activity. Depending on the nature of the scene, it provides suitable guidance and actions that can help users make informed decisions. For normal scenes, the system offers contextual information and location-related assistance, whereas for potentially hazardous situations it provides safety-oriented recommendations and reporting options.

The proposed system aims to combine geographical understanding and hazard awareness within a single platform, thereby improving image interpretation and assisting users in obtaining both location-based insights and safety-related information from visual data.

1.2 NEED AND RELEVANCE

Images are widely shared during travel, public events, accidents, disasters, and suspicious activities. A normal image may require contextual information about a place, while a risky image may require safe reporting guidance. Traditional image geolocation systems mainly focus on predicting a place, but practical users also need risk-aware interpretation and accountable action support.

The proposed system is relevant because it joins image understanding with

controlled decision support. It does not automatically notify authorities. Instead, it provides a human-in-the-loop flow where reporting is performed only after explicit user confirmation. This makes the system suitable for academic demonstration, public safety awareness, and image-based information retrieval.

1.3 MOTIVATION

With the rapid growth of social media and mobile photography, vast amounts of visual information are generated daily, often containing valuable clues about locations and environmental conditions. Most existing solutions focus solely on location identification or image description, lacking integrated hazard awareness. This project addresses this gap by developing a system that automatically analyzes images to provide both location insights and risk-aware guidance, enhancing situational awareness and supporting better decision-making in unfamiliar environments.

1.4 PROBLEM STATEMENT

To design and implement a multimodal web-based system that analyzes user-uploaded images, identifies possible location and scene context, detects hazard signals, and provides risk-aware actions with auditable user-confirmed reporting.

1.5 AIM AND OBJECTIVES

The aim of the project is to develop an interpretable multimodal image analysis system that combines geospatial reasoning with hazard-aware decision support.

Objectives:

- To integrate a vision-capable multimodal model for structured image analysis and location estimation.
- To implement deterministic risk-to-action mapping with human-confirmed reporting for safety-critical scenes.
- To persist sessions, model outputs, actions, and incidents for audit and evaluation.

1.6 PROJECT SCOPE AND LIMITATIONS

The project scope includes image upload, multimodal inference, structured response validation, confidence normalization, hazard-aware action mapping, learn-more lookup, optional map links, incident logging, and optional environment-gated WhatsApp notification after explicit confirmation.

The project does not include automatic emergency dispatch, legal identity verification, production-grade authentication, large-scale geotagged image retrieval, or forensic-level hazard verification. The current evaluation is based on a small project dataset and local run logs, so statistical conclusions are limited.

1.7 ORGANISATION OF THE REPORT

Chapter 1 introduces the project, motivation, problem statement, objectives, scope, and limitations. Chapter 2 presents the literature survey and research gap analysis. Chapter 3 describes the software requirement specification, use cases, data model, and functional model. Chapter 4 presents the project plan, estimates, risk management, and team organization. Chapter 5 explains system design, architecture, mathematical model, data design, and results. Chapter 6 presents advantages, limitations, and applications. Chapter 7 describes validation and testing. Chapter 8 concludes the report and presents future scope.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF EXISTING RESEARCH

Image geolocation and visual place recognition have been studied using retrieval-based methods, deep convolutional networks, and recent vision-language models. Early systems used local visual descriptors and large geotagged image collections for nearest-neighbor matching [1, 2]. Later approaches converted geolocation into classification over geographic regions or retrieval over learned place embeddings [3, 4]. Recent multimodal foundation models combine image understanding with natural language reasoning, allowing richer scene interpretation and more flexible user interaction [5, 6, 7, 8].

Hazard-aware visual decision support is related to explainable artificial intelligence, structured model outputs, safety policy design, and audit logging. In practical systems, location prediction alone is not enough. The system must also indicate uncertainty, provide rationale, and avoid automatic escalation without human confirmation. Retrieval-grounded and reasoning-based geolocation studies motivate stronger evidence linking, but most existing work still focuses on benchmark geolocation instead of incident-response workflow [9, 10, 11].

2.2 LITERATURE SURVEY

Table 2.1: Literature Survey

Sr. No	Paper Title	Journal / Conference	Authors & Year	Methodology	Research Gap
1	PlaNet - Photo Geolocation with Convolutional Neural Networks	ECCV	Weyand, Kostrikov and Philbin, 2016	CNN-based geolocation classification over geographic cells.	Strong geolocation focus, but limited explainability and no hazard-aware actions.

Sr. No	Paper Title	Journal / Conference	Authors & Year	Methodology	Research Gap
2	Revisiting IM2GPS in the Deep Learning Era	ICCV	Vo, Jacobs and Hays, 2017	Retrieval and deep feature based geolocation.	Depends on visual similarity datasets and does not provide decision-support workflow.
3	Learning Transferable Visual Models From Natural Language Supervision	ICML	Radford et al., 2021	Contrastive image-text pretraining for transferable visual representation.	Powerful visual-language alignment, but not directly designed for safety escalation.
4	PIGEON: Predicting Image Geolocations	CVPR	Haas, Skreta, Alberti and Finn, 2024	Multimodal geolocation using foundation model reasoning.	Geolocation performance is emphasized more than operational user actions.
5	Img2Loc: Revisiting Image Geolocalization using Multi-modality Foundation Models and Image-based Retrieval-Augmented Generation	SIGIR	Zhou et al., 2024	Foundation model plus retrieval-augmented image geolocation.	Retrieval grounding is useful, but incident logging and hazard response are outside scope.
6	LLMGeo: Benchmarking Large Language Models on Image Geolocation In-the-wild	arXiv	Wang et al., 2024	Benchmarking multimodal language models for geolocation.	Benchmarks model ability, but does not implement a complete safety-aware application.
7	Improving Image Geolocation with Multimodal Deep Learning	ICONIP	Sinski, Zychowski and Mandziuk, 2024	Uses multimodal deep learning for image geolocation improvement.	Focuses on geolocation improvement, not user-confirmed hazard reporting.

Sr. No	Paper Title	Journal / Conference	Authors & Year	Methodology	Research Gap
8	IM2City: Image Geo-localization via Multi-modal Learning	ACM SIGSPA-TIAL Workshop	Wu and Huang, 2022	Combines image and auxiliary modality cues for city-scale localization.	City-level localization does not directly support safety triage actions.
9	Tell Me Where You Are: Multimodal LLMs Meet Place Recognition	arXiv	Lyu et al., 2024	Evaluates multimodal LLMs for place recognition and contextual location reasoning.	Does not include incident persistence or notification policy.
10	BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	ICML	Li et al., 2023	Aligns frozen image encoders with large language models using lightweight querying.	General vision-language method without domain-specific hazard action mapping.
11	VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization	arXiv	Waheed et al., 2025	Uses VLM guidance for large-scale visual place recognition.	Emphasizes retrieval quality, not controlled user-facing report workflow.
12	From Pixels to Places: A Systematic Benchmark for Evaluating Image Geolocation Ability in Large Language Models	arXiv	Li et al., 2025	Benchmarks LLM geolocation ability across image-to-place tasks.	Evaluation is benchmark-oriented and does not measure incident action outcomes.
13	GeoChain: Multimodal Chain-of-Thought for Geographic Reasoning	EMNLP Findings	Yerramilli et al., 2025	Decomposes geographic reasoning using multimodal chain-of-thought.	Reasoning improves explainability, but operational safety logging is outside scope.

Sr. No	Paper Title	Journal / Conference	Authors & Year	Methodology	Research Gap
14	GeoGuess: Multi-modal Reasoning based on Hierarchy of Visual Information in Street View	arXiv	Cheng et al., 2025	Uses hierarchical visual information for street-view geolocation reasoning.	Street-view reasoning is not directly integrated with emergency response actions.
15	Distinctive Image Features from Scale-Invariant Keypoints	IJCV	Lowe, 2004	Introduces SIFT local features for robust image matching.	Classical matching lacks semantic risk understanding and natural language rationale.
16	Visual Place Recognition with Repetitive Structures	IEEE TPAMI	Torii et al., 2015	Improves place recognition under repeated urban structures and viewpoint changes.	Robust place matching does not classify hazard severity or confidence-based action paths.
17	NetVLAD: CNN Architecture for Weakly Supervised Place Recognition	CVPR	Arandjelovic et al., 2016	Learns compact place descriptors using weak supervision.	Retrieval descriptors alone do not provide explainable safety decisions.
18	SuperGlue: Learning Feature Matching with Graph Neural Networks	CVPR	Sarlin et al., 2020	Learns graph-based feature matching for reliable correspondences.	Useful for visual evidence but not a complete multimodal decision-support system.
19	Visual Instruction Tuning	NeurIPS	Liu et al., 2024	Trains instruction-following vision-language assistants.	General assistant behavior still requires project-specific validation and policy control.

Sr. No	Paper Title	Journal / Conference	Authors & Year	Methodology	Research Gap
20	Gemini: A Family of Highly Capable Multimodal Models	arXiv	Gemini Team, 2024	Describes a family of multimodal models for text, image, and reasoning tasks.	Model capability alone does not define auditable incident workflow or local system policy.

2.3 LITERATURE SURVEY DESCRIPTION

The literature survey covers image geolocation and hazard-aware visual decision support across classical feature matching, deep convolutional networks, and recent vision-language models. From the surveyed work, five papers are most relevant to this project and are described below:

- **CLIP (Radford et al., 2021)** [5] aligns images and text by contrastive pre-training on large image-text pairs, enabling flexible language-guided visual understanding. Its limitation is that the generalist design has no domain-specific tuning for safety-critical decisions.
- **PIGEON (Haas et al., 2024)** [12] uses foundation-model reasoning to produce structured geographic predictions with confidence scores from images. Its limitation is that it targets prediction accuracy only and does not provide human-confirmed reporting or audit logging.
- **Img2Loc (Zhou et al., 2024)** [9] combines a foundation model with retrieval-augmented generation to ground geolocation in similar reference images for better robustness. Its limitation is that it emphasizes retrieval accuracy without safety triage or incident logging.
- **GeoChain (Yerramilli et al., 2025)** [11] applies multimodal chain-of-thought to decompose geographic reasoning into explicit, transparent steps. Its limitation is that it improves explainability but does not integrate operational policy

or incident reporting.

- **Gemini (Gemini Team, 2024)** [8] is a family of multimodal models with strong reasoning and structured-output support, suitable as the inference engine for this project. Its limitation is that raw model capability still requires external safeguards such as confirmation, logging, and policy mapping.

Overall, the survey shows that while existing work performs well on geolocation and visual understanding benchmarks, most approaches do not integrate hazard-aware decision support, human-confirmed reporting, and audit logging into a single workflow.

2.4 RESEARCH GAP ANALYSIS

The reviewed literature shows significant progress in image geolocation, visual place recognition, and vision-language reasoning. However, most approaches are evaluated using location accuracy, retrieval performance, or benchmark scores. In real-world use, a user may need both place information and risk-aware guidance.

The main gaps identified are:

- Lack of combined geolocation and hazard-aware response in a single user-facing workflow.
- Limited use of structured output validation for model responses.
- Limited focus on deterministic action mapping from risk level and confidence.
- Lack of human-confirmed incident reporting and audit logging in geolocation prototypes.
- Limited confidence calibration and conservative fallback design for ambiguous images.

The proposed project addresses these gaps by combining multimodal image analysis, structured JSON output, deterministic policy rules, action buttons, and persistent logs for evaluation.

CHAPTER 3

SOFTWARE REQUIREMENT

SPECIFICATION

3.1 INTRODUCTION

3.1.1 Purpose and Scope of Document

Facilitating other Documentation: This SRS serves as a foundational reference for all future project documentation, including system design documents, test plans, user manuals, and project reports. It ensures consistency across development phases by providing a clear understanding of system requirements and expected behavior.

Product Validation: The SRS enables validation by establishing measurable criteria for evaluating the system's performance and verifying that the final implementation meets user requirements. It acts as a benchmark for acceptance testing to ensure the predicted geolocation results are accurate, reliable, and aligned with project objectives.

3.1.2 Characteristics of a Software Requirement Specification

Accuracy: The SRS clearly defines the functional and non-functional requirements without ambiguity. Every requirement accurately reflects the intended system behavior, such as input image handling, machine learning-based geolocation prediction, and result interpretation. Accurate details ensure developers and stakeholders share a common understanding.

Completeness: The SRS provides a complete description of all required system features, constraints, performance expectations, and interfaces. It includes functional requirements such as dataset processing, model training, and location prediction, as well as non-functional aspects like performance, security, usability, and scalability. No essential requirement is omitted.

Prioritization of Requirements: The requirements in the SRS are organized based on importance and implementation priority. Critical functionalities such as image preprocessing and model prediction are placed at higher priority, followed by enhancements like user interface improvements, accuracy tuning, and dataset expansion. Prioritization ensures efficient planning and phased development.

3.1.3 Overview of Responsibilities of Developer

The development responsibilities include frontend design, backend API implementation, multimodal model integration, database schema design, validation logic, safety policy implementation, and testing. The developer must ensure that the system does not perform automatic authority dispatch and that all report actions are explicitly confirmed by the user.

3.2 USAGE SCENARIO

The system is designed for users who want to understand an image, identify possible place context, or receive safety-aware guidance. A user opens the web application, uploads an image, and submits it for analysis. The backend sends the image and prompt to the multimodal model, validates the returned JSON, applies policy rules, saves logs, and returns a response with actions.

3.2.1 User Profiles

Table 3.1: System Entities Description

Sr. No.	Actor	Description
1	General User	Uploads an image and receives scene explanation, location estimate, risk level, and actions.
2	Public Safety Observer	Uses the report flow when a hazardous or suspicious image is detected and confirms incident logging.
3	Reviewer / Evaluator	Reviews stored model outputs, actions, confidence values, and incident records for project evaluation.
4	System Administrator	Configures API keys, database path, upload size, and optional WhatsApp notification settings.

3.2.2 Use Cases

Table 3.2: Use Cases

Sr. No.	Use Case	Description	Actors	Assumption
1	Analyze Image	User uploads an image and receives a structured scene interpretation.	User	Image format is supported.
2	Learn More	User selects learn-more action for a normal landmark or place.	User	Place/entity is detected.
3	Open Map	User opens a map search for the detected location.	User	Location guess is available.
4	View Explanation	User checks why a result was produced.	User	Rationale is returned or fallback rationale exists.
5	Report Suspicious Activity	User confirms incident reporting for high-risk output.	User	Confirmation is required before logging.
6	Review Logs	Evaluator checks sessions, actions, and incidents.	Reviewer	Database records are available.

3.2.3 Use Case View

The use case view represents interaction between users and the multimodal assistant. The primary user uploads an image and receives analysis. Based on the response, the user may learn more, open a map, view rationale, or confirm a report. The reviewer interacts with logs for evaluation.

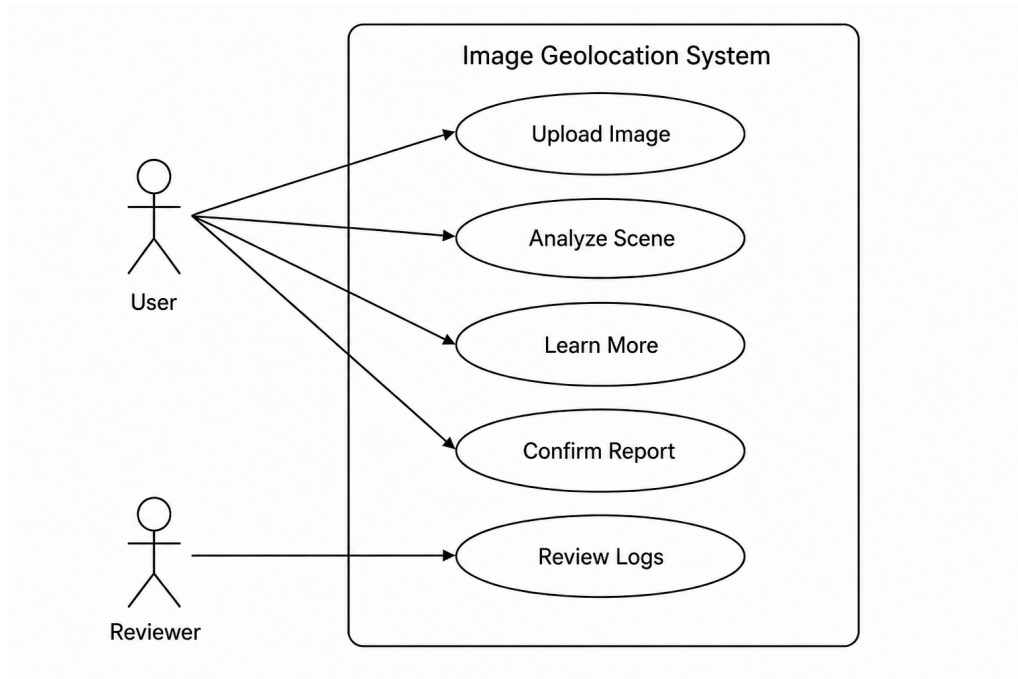


Figure 3.1: Use case diagram

3.3 DATA MODEL AND DESCRIPTION

3.3.1 Data Description

The system uses SQLite as the persistence layer. It stores session data, messages, structured model outputs, user actions, and incident records. Raw images are not stored by default; an image hash is saved for traceability and privacy.

3.3.2 Data Objects and Relationships

The main database tables are:

- **sessions:** Stores conversation sessions.
- **messages:** Stores user and assistant messages with optional image hash.
- **model.outputs:** Stores risk level, confidence, and full JSON payload.
- **actions:** Stores user-selected actions such as learn-more and report.
- **incidents:** Stores user-confirmed incident reports and optional notification status.

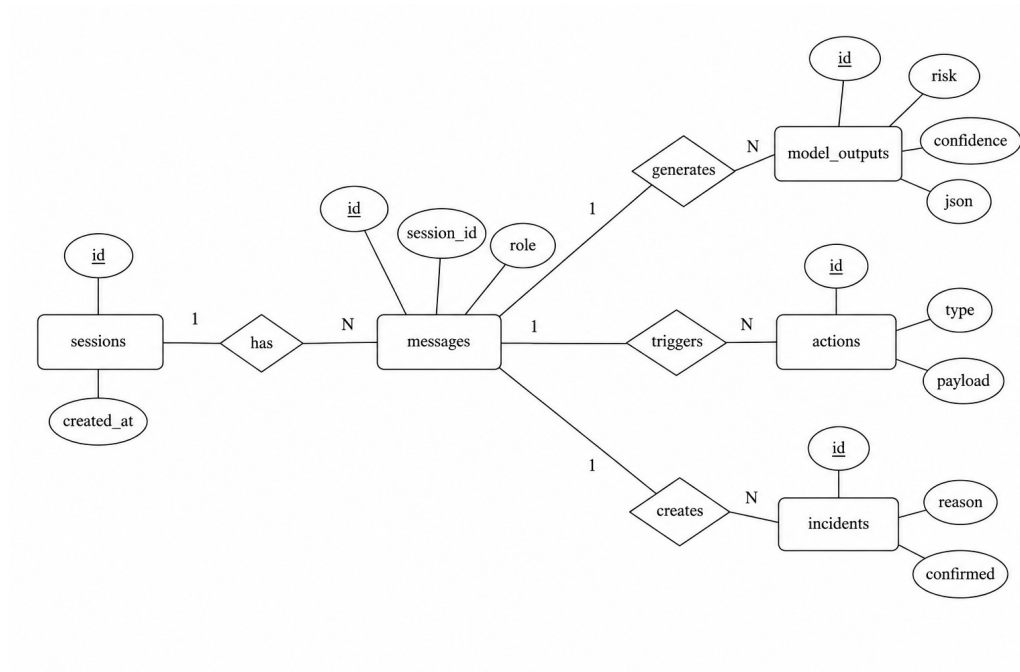


Figure 3.2: ER diagram

3.4 FUNCTIONAL MODEL AND DESCRIPTION

3.4.1 Data Flow Diagram

The data flow starts when a user submits an image. The frontend sends a multipart request to the backend. The backend validates input, hashes the image, invokes Gemini Vision, parses the model response, applies policy rules, saves records, and returns a response to the frontend.

3.4.1.1 Level 0 Data Flow Diagram

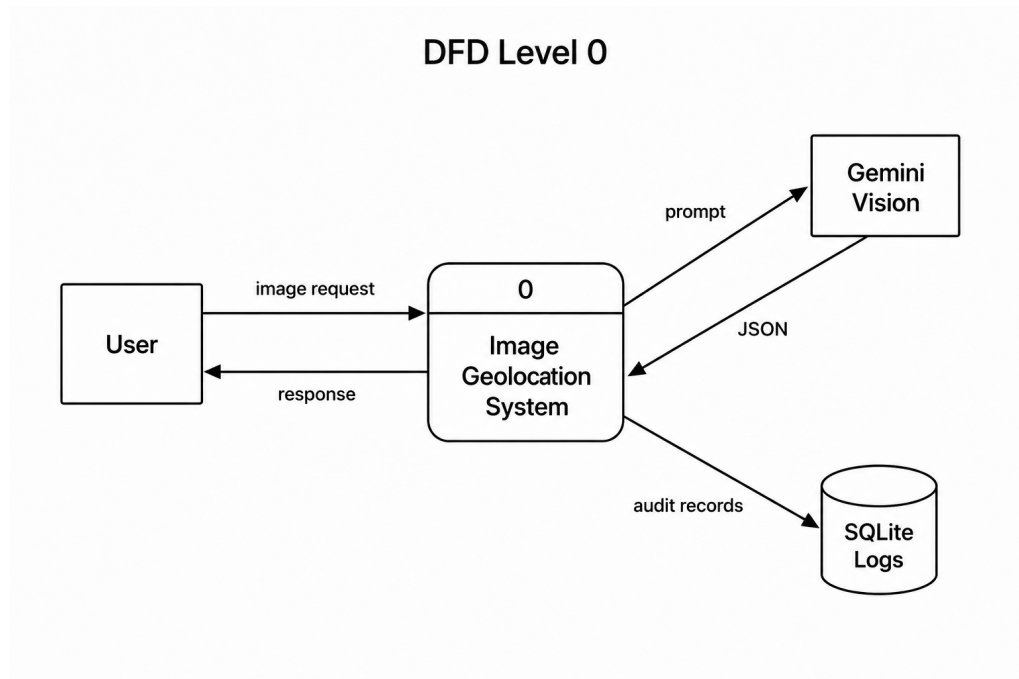


Figure 3.3: DFD Level 0 diagram

3.4.1.2 Level 1 Data Flow Diagram

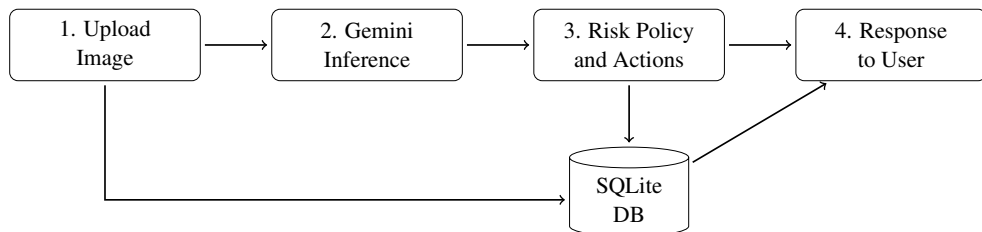


Figure 3.4: DFD Level 1 diagram

3.4.2 Description of Functions

The major functions of the system are image upload, model inference, JSON extraction, JSON repair, output normalization, hazard signal detection, action generation, learn-more lookup, map link generation, incident reporting, and audit logging.

3.4.3 Activity Diagram

The activity diagram represents the end-to-end workflow from image upload to action selection.

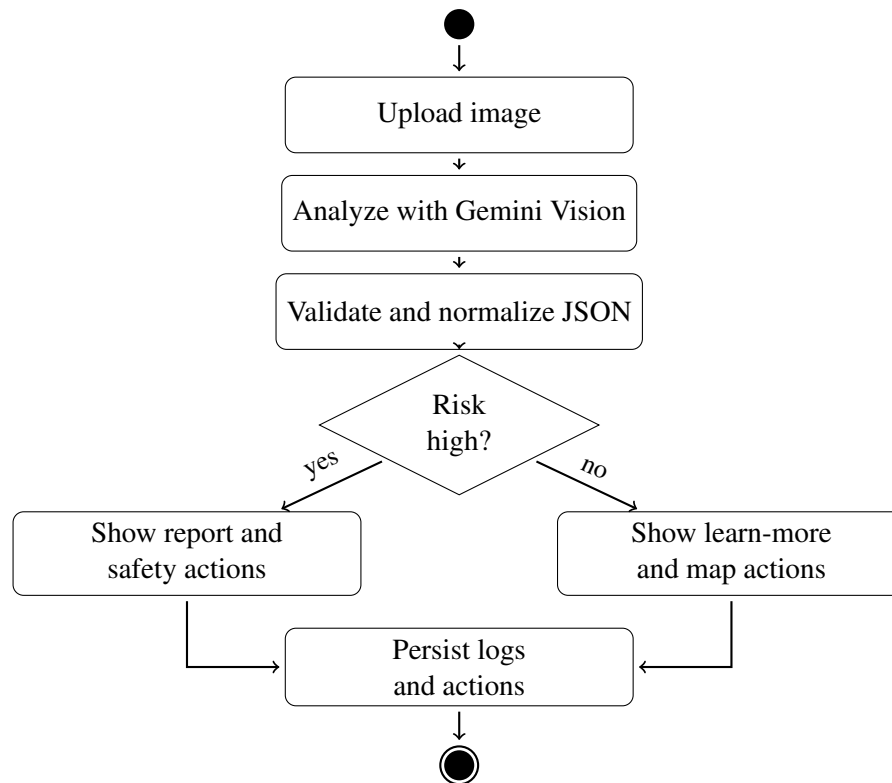


Figure 3.5: Activity diagram

3.4.4 Non Functional Requirements

- **Performance:** The MVP targets response latency below 8 seconds for common demo cases.
- **Reliability:** Invalid model output must trigger repair or safe fallback.
- **Security:** API keys must be stored in environment variables and not logged.
- **Privacy:** Raw images are not stored by default; image hashes are persisted.
- **Usability:** The interface must show image preview, chat messages, and inline action buttons.
- **Auditability:** Every model output, action, and incident must be stored for review.

3.4.5 State Diagram

The system state changes from image selection to analysis, output generation, action rendering, and optional incident creation.

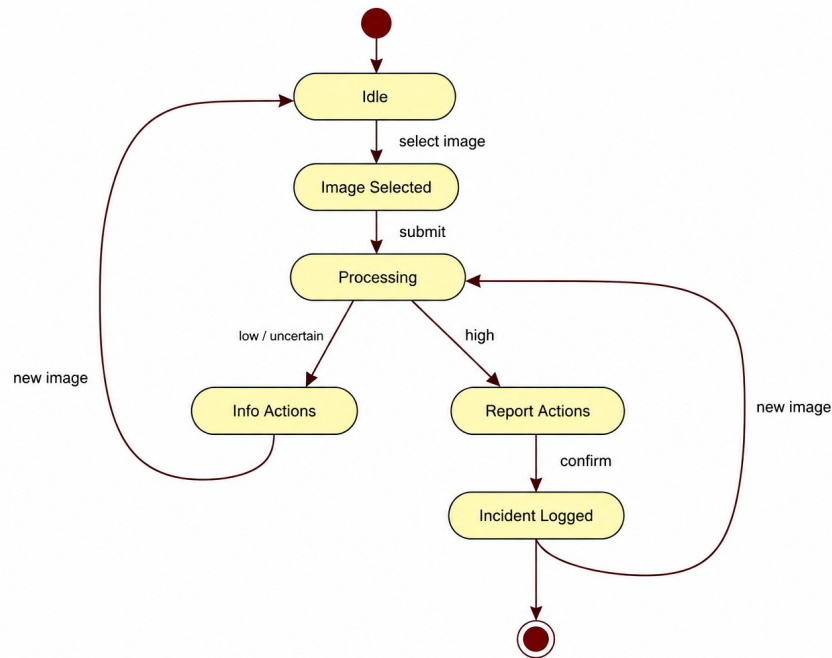


Figure 3.6: State transition diagram

3.4.6 Design Constraints

The system must not perform automatic external authority dispatch. WhatsApp notification is optional, environment-gated, and requires explicit user confirmation. The model must return strict JSON where possible, and fallback behavior must be conservative when confidence is low or output is invalid.

3.4.7 Software Interface Description

The backend exposes REST APIs for analysis, learn-more, report, and health check. The frontend communicates with these APIs using HTTP requests. Gemini Vision is accessed through the Google Generative Language API. Wikipedia is used for learn-more summaries, and Google Maps search links are generated for detected locations.

CHAPTER 4

PROJECT PLAN

4.1 PROJECT ESTIMATES

The project was planned using an iterative software development approach. The work was divided into requirement analysis, literature survey, design, implementation, integration, testing, and documentation.

1. **Requirement Gathering:** The team studied the need for image geolocation, scene explanation, risk triage, user-confirmed reporting, and audit logging.
2. **Analysis:** Functional requirements, non-functional requirements, user roles, model constraints, and safety requirements were identified.
3. **Design:** The team designed the system architecture, data model, API flow, UI flow, and risk-to-action policy.
4. **Implementation:** The backend, frontend, database schema, model integration, action handlers, and incident logging were implemented.
5. **Testing:** Normal landmark images, hazardous images, ambiguous cases, learn-more actions, and report flows were tested.
6. **Documentation:** The final report, research paper, diagrams, results, limitations, and future scope were prepared.

4.1.1 Reconciled Estimates

4.1.1.1 Cost Estimate

The project uses open-source development tools and academic infrastructure. The main cost components are internet usage, development machines, optional API usage, and documentation/printing. Since the prototype runs locally and uses SQLite, infrastructure cost is minimal.

Table 4.1: Cost Estimate

Sr. No.	Resource	Estimated Cost
1	Development laptops and college laboratory facilities	Existing resources
2	Open-source software tools	No direct cost
3	Internet and API access for model testing	Variable / demo usage
4	Report printing and binding	As per college requirement

4.1.1.2 Time Estimates

Table 4.2: Time Estimate

Sr. No.	Activity	Duration
1	Requirement analysis and literature survey	3 weeks
2	System design and database design	2 weeks
3	Backend and model integration	4 weeks
4	Frontend development and action workflow	4 weeks
5	Testing, metrics, and report preparation	3 weeks

4.1.2 Project Resources

- **People:** Four student developers, internal guide, project coordinator, and reviewers.
- **Hardware:** Development laptops with internet connection.
- **Software:** Python, Flask, SQLite, React, Vite, TypeScript, Tailwind CSS, Gemini Vision API, Wikipedia package, LaTeX.
- **Tools:** VS Code, Git, browser developer tools, draw.io/diagram tools, pdflatex.

4.2 RISK MANAGEMENT W.R.T. NP HARD ANALYSIS

The project is not an NP-hard optimization problem in the classical computational sense. However, it contains uncertainty and risk because visual scene interpretation, geolocation, and hazard detection are open-ended AI tasks. Risk management therefore focuses on model uncertainty, fallback behavior, API failure, privacy, and false alerts.

4.2.1 Risk Identification

- Model returns invalid or incomplete JSON.
- Model produces incorrect location or risk level.
- Hazard keyword fallback creates false alerts.
- API key or network failure prevents model inference.

- User reports an incident without proper context.
- Raw image storage may create privacy concerns.

4.2.2 Risk Analysis

Table 4.3: Risk Table

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	Invalid model output or parsing failure	Medium	Low	High	High
2	Incorrect risk classification	Medium	Low	High	High
3	Gemini API unavailable or key missing	Low	Medium	Medium	Medium
4	False incident report by user	Low	Low	High	High
5	Privacy concern due to image handling	Low	Low	High	High

4.2.3 Overview of Risk Mitigation, Monitoring, Management

Table 4.4: Risk Mitigation Plan

Risk ID	Mitigation Strategy	Monitoring Method
1	JSON extraction, one repair pass, and safe fallback response	Check model output logs
2	Deterministic confidence thresholds and hazard policy rules	Review risk distribution and sample outputs
3	Clear backend error handling and fallback messages	Monitor API response errors
4	Require explicit user confirmation before incident creation	Review incident table
5	Store image hash instead of raw image by default	Inspect message records and configuration

4.3 PROJECT SCHEDULE

4.3.1 Project Task Set

- Task 1: Domain finalization and literature survey.

- Task 2: Requirement analysis and SRS preparation.
- Task 3: UI, API, database, and policy design.
- Task 4: Backend implementation and Gemini integration.
- Task 5: Frontend implementation and action buttons.
- Task 6: Testing with normal, hazardous, and ambiguous images.
- Task 7: Metrics analysis, report writing, and paper preparation.

4.3.2 Task Network

The task network follows a dependency chain: literature survey and requirements lead to architecture design; architecture design leads to backend and frontend implementation; implementation leads to testing; testing leads to result analysis and documentation.

4.3.3 Timeline Chart

Table 4.5: Timeline Chart

Sr. No.	Task Name	Start Month	End Month
1	Domain study and base paper selection	June 2025	July 2025
2	Requirement analysis and synopsis	August 2025	September 2025
3	System design and diagrams	September 2025	October 2025
4	Backend and database implementation	November 2025	January 2026
5	Frontend and action workflow implementation	January 2026	February 2026
6	Testing, evaluation, and report preparation	February 2026	April 2026
7	Final documentation and submission	April 2026	May 2026

4.4 TEAM ORGANIZATION

4.4.1 Team Structure

The team consists of four student members working under the guidance of the internal guide. Responsibilities were distributed across research, backend, frontend, database, testing, documentation, and presentation.

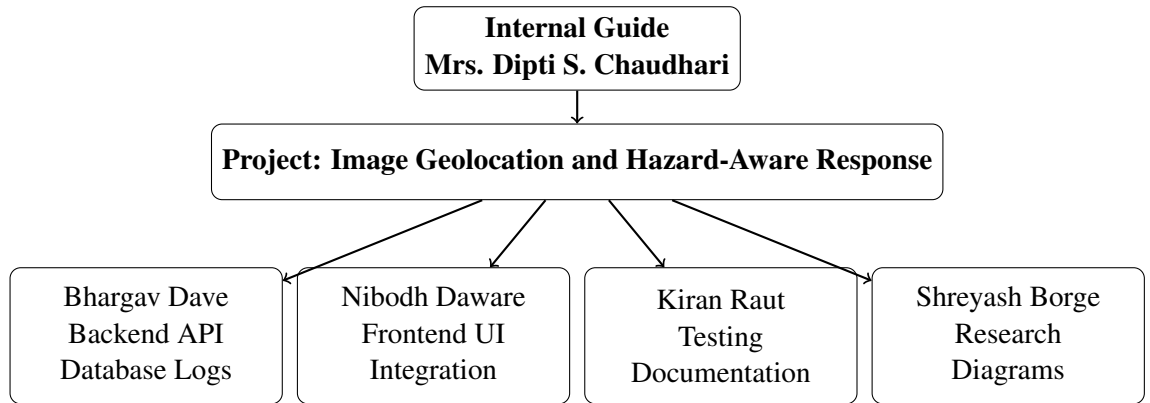


Figure 4.1: Team structure

4.4.2 Management Reporting and Communication

The team followed regular internal discussions, guide reviews, project lab sessions, and milestone-based reporting. Progress was tracked through working demonstrations, report updates, and review feedback.

Table 4.6: Management Plan

Sr. No.	Month	Description
1	June-July	Domain discussion, paper search, and problem finalization.
2	August-September	Requirement analysis, synopsis, and preliminary design.
3	October-November	Architecture, database design, and backend planning.
4	December-February	Implementation of backend, frontend, and action workflow.
5	March-April	Testing, evaluation, metrics, and report drafting.
6	May	Final report completion and submission preparation.

CHAPTER 5

SYSTEM DESIGN

5.1 INTRODUCTION

System design converts the requirements into a working architecture. The proposed system is designed as a client-server web application with a React frontend, Flask backend, SQLite database, and Gemini Vision model integration. The architecture separates user interaction, model inference, policy logic, action handling, and persistence.

5.2 SYSTEM ARCHITECTURE DESIGN

The system architecture is shown in Fig. 5.1. The frontend accepts image input and displays chat messages. The backend receives the request, validates input, sends the image to Gemini Vision, parses the response, applies normalization and risk rules, saves records, and returns action buttons to the frontend.

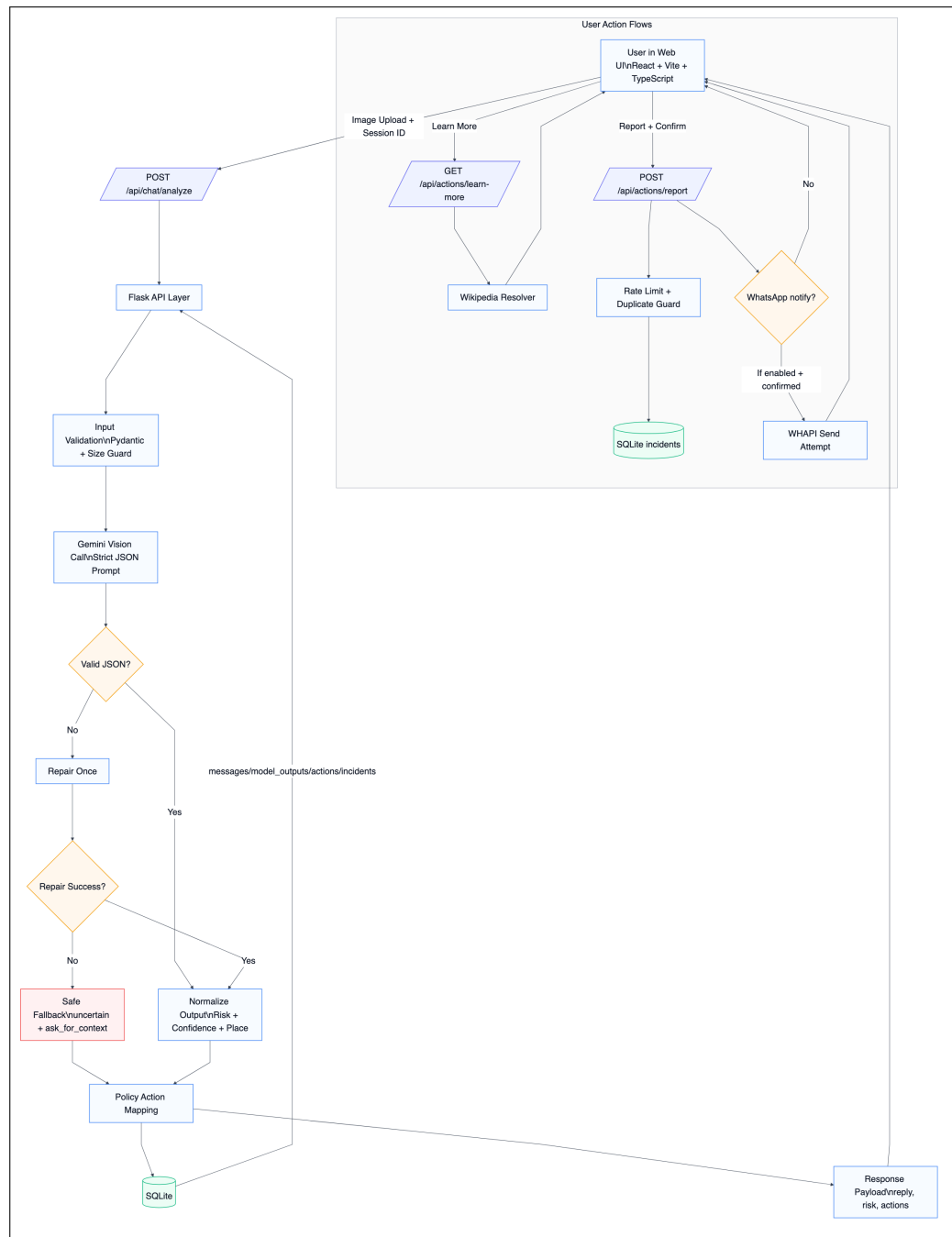


Figure 5.1: System architecture

The main modules are:

- **User Interface:** Provides image upload, preview, chat timeline, and action buttons.
- **Backend API:** Handles analyze, learn-more, report, and health endpoints.
- **Multimodal Inference:** Sends image and prompt to Gemini Vision and re-

quests strict JSON.

- **Parser and Normalizer:** Extracts JSON, repairs invalid output once, validates fields, and normalizes confidence.
- **Risk Policy:** Converts model output and hazard signals into final risk and confidence.
- **Action Mapper:** Generates actions such as report, learn-more, open map, safety tips, and why.
- **Persistence Layer:** Stores sessions, messages, model outputs, actions, and incidents in SQLite.

5.3 MATHEMATICAL MODEL FOR THE PROPOSED SYSTEM

Let the input image be represented by I and optional user context be represented by T . The multimodal model produces a structured output:

$$o = M(I, T)$$

where o contains summary, risk level, confidence, entities, possible location, rationale, and recommended actions.

Let r_m be the model risk level and c_m be the model confidence. Let $h = 1$ if hazard keywords are present in the summary, rationale, user context, or location text; otherwise $h = 0$. The final risk is:

$$r = \begin{cases} \text{high}, & h = 1 \text{ and } r_m \in \{\text{low}, \text{uncertain}\} \\ r_m, & \text{otherwise} \end{cases}$$

The confidence value is normalized to the range $[0, 1]$. When a hazard signal is detected, the confidence is adjusted as:

$$c = \max(c_m, 0.72) \quad \text{when } h = 1$$

The action set A is selected using deterministic policy rules:

$$A = \begin{cases} \{\text{ask_for_context, why}\}, & c < 0.45 \text{ or } r = \text{uncertain} \\ \{\text{report, safety_tips, why}\}, & r = \text{high} \\ \{\text{safety_tips, why}\}, & r = \text{medium} \\ \{\text{learn_more, open_map, why}\}, & r = \text{low} \end{cases}$$

Algorithm 1: Hazard-Aware Image Analysis Flow

Input: Input image I and optional message T

Output: Structured response and action set A

Generate structured model output $o \leftarrow M(I, T)$;

Extract JSON from model output;

if *JSON is invalid* **then**

 Attempt one repair pass;

if *repair fails* **then**

 Return safe fallback output;

Normalize summary, risk, confidence, entities, and location;

Detect hazard signal h ;

Apply risk and confidence policy;

Generate action set A ;

Persist message, output, and action records;

Return response to frontend;

5.4 DATA DESIGN

The internal data structure uses Python dictionaries and Pydantic models for API validation. The global data structure is the SQLite database. Temporary data includes uploaded image bytes, base64 encoded image content, parsed model JSON, and generated action payloads.

5.4.1 Database Description

Table 5.1: Database Tables

Sr. No.	Table	Purpose
1	sessions	Stores unique user sessions and creation time.
2	messages	Stores user and assistant messages with image hash.
3	model_outputs	Stores risk level, confidence, JSON payload, and timestamp.
4	actions	Stores user action type and payload.
5	incidents	Stores confirmed reports and notification status.

5.4.2 Component Design

Table 5.2: Component Design

Sr. No.	Component	Responsibility
1	React UI	Image upload, preview, chat display, and action button rendering.
2	Flask API	Request handling, validation, response construction, and database operations.
3	Gemini Integration	Multimodal image analysis and structured output generation.
4	Policy Engine	Risk normalization, hazard keyword escalation, and action mapping.
5	SQLite Store	Persistent audit logs for sessions, messages, outputs, actions, and incidents.

5.5 RESULTS AND DISCUSSION

The project evaluation uses available run logs. A run is one invocation of the image-analysis endpoint that produces one structured model output. The available evaluation set contains 22 runs across 14 sessions.

Table 5.3: Available Dataset and Log Summary

Measure	Value
Total analyzed runs	22
Distinct sessions	14
Low-risk outputs	8
High-risk outputs	8
Uncertain outputs	6
Average confidence	0.623
Confidence range	0.00–0.95

The reported confidence values are model confidence and policy-normalized confidence values, not independent expert accuracy labels. For the manually checked Shaniwar Wada case study, the system produced the expected action path for both samples: the regular image was classified as low risk with a learn-more action, and the fire-affected image was classified as high risk with a report action. Therefore, the verified case-study risk-action accuracy is 2 out of 2, or 100%. A larger labelled test set is required to calculate precision, recall, and full hazard-classification accuracy.

Table 5.4: Hazard-Action Outcome Summary

Outcome	Count
User-confirmed incidents	10
Environment-enabled notification attempts	3
Successful notification sends	3
Local-only incident records	7
Report actions logged	10
Learn-more actions logged	12

Incident reporting and external notification are treated as separate stages. All ten incidents were user-confirmed and stored in SQLite. Only three of those reports had external notification enabled by the environment and user flow; all three enabled attempts were sent successfully.



(a) Regular Shaniwar Wada image



(b) Hazardous Shaniwar Wada image

Figure 5.2: Test images used for case study

For the normal Shaniwar Wada image, the stored output produced low risk with confidence 0.95 and the primary action was Learn More. For the fire-affected Shaniwar Wada image, the stored output produced high risk with confidence 0.85 and the primary action was Report. This demonstrates that the same place can receive different action paths depending on visual hazard evidence.

CHAPTER 6

OTHER SPECIFICATIONS

6.1 ADVANTAGES OF THE SYSTEM

- **Multimodal understanding:** The system can analyze image content along with user context.
- **Structured output:** Model responses are converted into auditable fields instead of unstructured text only.
- **Risk-aware actions:** The response changes based on risk level and confidence.
- **Human-in-the-loop safety:** Incident reporting requires explicit confirmation from the user.
- **Audit trail:** Sessions, messages, model outputs, actions, and incident records are persisted.
- **Privacy-conscious storage:** Raw image storage is avoided by default and image hashes are stored.
- **Extensible design:** The architecture can support additional models, retrieval systems, and notification providers in future.

6.2 LIMITATIONS OF THE SYSTEM

- The evaluation dataset is small and limited to available project logs.
- The system depends on the accuracy and availability of the multimodal model.
- Hazard detection is not equivalent to legal or forensic verification.
- Confidence scores are model-provided and require stronger calibration for production use.
- The current implementation does not use vector database retrieval for geo-tagged image grounding.
- Hazard demonstrations are mainly fire-related, while other emergency types need more testing.

- The system does not include production-grade authentication, role-based access control, or legal compliance workflows.

6.3 APPLICATIONS OF THE SYSTEM

- Tourist landmark explanation and learn-more assistance.
- Public safety awareness and guided incident reporting.
- Academic demonstration of multimodal LLM-based decision support.
- Image-based context retrieval for educational use.
- Smart city monitoring support with human verification.
- Disaster awareness prototypes for fire, flood, collapse, or other visible hazards.
- Digital verification workflows where image context and explanation are required.

CHAPTER 7

VALIDATION AND TESTING

7.1 TESTING METHODOLOGY

Testing was performed using functional testing, integration testing, UI testing, API testing, and log-based result validation. The focus was to verify that the system accepts images, returns structured responses, handles invalid model output safely, maps risk to correct actions, and persists records.

The testing flow was:

1. Upload normal landmark image.
2. Verify low-risk response and learn-more action.
3. Upload hazardous image.
4. Verify high-risk response and report action.
5. Test ambiguous or invalid response behavior.
6. Verify database records for messages, outputs, actions, and incidents.
7. Verify report confirmation and optional notification status.

7.2 TEST CASES

Table 7.1: Test Case Summary

Sr. No	Test Case Title / Description	Testing Methodology Applied	Result	Defects Identified
1	Upload supported image file and submit for analysis	Functional testing	PASS	None
2	Analyze regular Shaniwar Wada image	System testing	PASS	None
3	Analyze fire-affected Shaniwar Wada image	System testing	PASS	None
4	Verify high-risk output shows Report action	Black-box testing	PASS	None
5	Verify low-risk output shows Learn More and Open Map actions	Black-box testing	PASS	None
6	Verify uncertain or low-confidence response asks for more context	Boundary testing	PASS	None
7	Verify report action requires confirmation before incident creation	Functional testing	PASS	None
8	Verify model output, action, and incident records are saved in SQLite	Integration testing	PASS	None
9	Verify missing/invalid model response returns safe fallback	Negative testing	PASS	None
10	Verify learn-more action fetches external knowledge summary	Integration testing	PASS	Depends on Wikipedia availability

7.3 VALIDATION OF RESULTS

The stored logs show 22 analyzed runs across 14 sessions. The risk distribution includes 8 low-risk outputs, 8 high-risk outputs, and 6 uncertain outputs. Confidence values range from 0.00 to 0.95 with an average of 0.623.

Table 7.2: Confidence Band Distribution

Band	Count	Share
Low (< 0.30)	6	27.3%
Mid (0.30–0.79)	5	22.7%
High (≥ 0.80)	11	50.0%

The case-study images validate the main intended behavior: the normal image leads to an informational action, while the hazardous image leads to a report-oriented action. The system therefore satisfies the MVP acceptance criteria for normal image interpretation, risky image escalation, ambiguity handling, and incident log review.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

8.1 CONCLUSION

This project successfully developed a multimodal image understanding system that integrates image geolocation, risk assessment, and guided action support. The system provides informational actions for normal scenes and safety-oriented actions for high-risk scenes through a human-confirmed, auditable reporting flow. The implementation demonstrates how visual analysis combined with natural language reasoning can support structured, accountable decision-making. Evaluation across 22 test runs shows balanced detection capabilities with proper risk classification, confirming the viability of multimodal approaches for incident response support. The system successfully addresses the core objectives of providing location estimates, hazard detection, and user-confirmed reporting while maintaining comprehensive audit logs for transparency and accountability.

8.2 FUTURE SCOPE

Future enhancements could focus on improving the system's scalability and accuracy. First, implementing vector retrieval and embedding-based search would enable better geospatial accuracy and faster incident matching across large datasets. Second, expanding the system with role-based access control, expert incident review workflows, and real-time dashboard analytics would enhance operational usability for safety teams and enable comprehensive incident analysis and trend detection.

CHAPTER 9

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